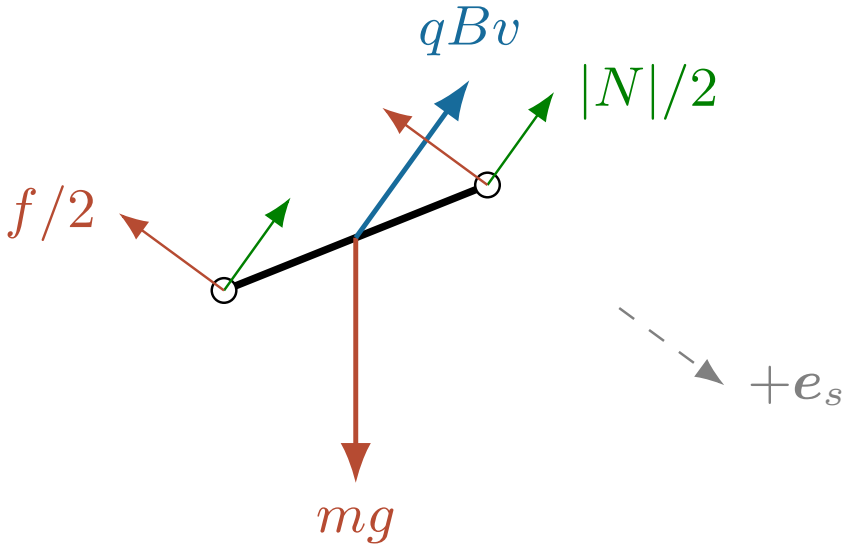
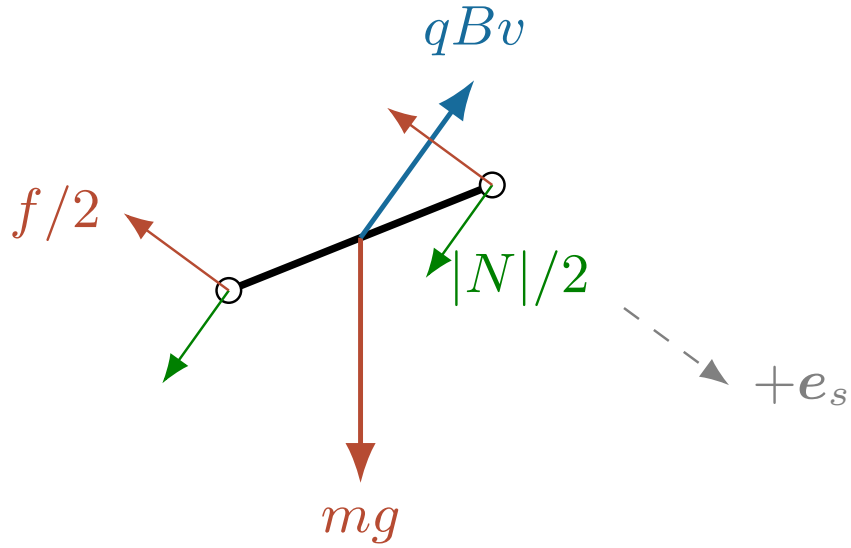
$v < v_c$



$$N = mg\cos\theta - qBv$$

$v > v_c$



$$N = mg\cos\theta - qBv$$